OPERATIONS & INFRASTRUCTURE GUIDE

MediaHub: Synology NAS Database Migration

A complete, step-by-step production handbook for consolidating MediaHub's Cloudflare and Vercel deployments onto a unified, self-hosted PostgreSQL database on a Synology NAS.

Domain: nushmedia.app

Target Engine: PostgreSQL 16 (Docker)

Network Security: Cloudflare Zero Trust Tunnel

Target NAS: Synology DSM 7.2+

1. Architecture Overview

Currently, MediaHub runs across two deployments: the production site on Cloudflare Workers (`nushmedia.app`, using D1 SQLite) and a preview/staging environment on Vercel (using Neon PostgreSQL).

By moving to a **Synology NAS**, both deployments (plus any local laptop kiosks) connect to a **single, centralized PostgreSQL database** running on your school/club hardware.

Component	Before Migration	After Migration (Synology NAS)
Database Engine	Cloudflare D1 (SQLite) + Neon PostgreSQL	PostgreSQL 16 (Unified)
Storage & Ownership	Cloud-hosted, isolated data silos	Local NAS RAID Pool (Encrypted & Backed Up)
Public Domain	`nushmedia.app` on Cloudflare Workers	`nushmedia.app` pointing to Synology via Cloudflare Tunnel
School Firewall	N/A (Hosted in Cloud)	Zero Open Ports Needed (Outbound Tunnel)
Database Latency	30-80ms cloud round-trip	< **1ms** for local kiosks; fast edge cached for web

> **Key Security Concept:** Your school NAS sits behind an internal firewall with a private IP (e.g. `192.168.1.x`). School IT will never permit opening public database ports. We solve this using **Cloudflare Tunnel (cloudflared)**, which establishes an encrypted *outbound-only* connection from the NAS to Cloudflare, requiring **zero open router ports**.

2. Prerequisites & Requirements

- **Synology NAS:** Running DSM 7.2 or later with **Container Manager** installed from the Package Center.
- **Cloudflare Zero Trust Account:** Free tier, associated with your active domain `nushmedia.app`.
- **Storage Space:** At least 2 GB free on your primary storage pool (e.g. `/volume1`).
- **Access Rights:** Synology DSM administrator account and Cloudflare dashboard access.

3. Step-by-Step Implementation

1 Prepare NAS Shared Folder for Persistent Storage

To ensure database records, audit sessions, and user profiles survive container restarts and DSM upgrades, database files must be stored directly on the NAS storage pool.

1. Log in to Synology DSM web portal (`http://<NAS_IP>:5000`).
2. Open **File Station**.
3. Under the **docker** shared folder (created when Container Manager was installed), create the following folder hierarchy:
   `/volume1/docker/mediahub/pgdata`
4. Right-click the `mediahub` folder > **Properties** > **Permissions** > verify your admin account has full Read & Write privileges.

2 Deploy PostgreSQL 16 via Container Manager

Use Synology's built-in Docker Compose engine (Projects) to run PostgreSQL with optimized health checks and automatic restarts.

1. Open **Container Manager** on your Synology NAS.
2. Click **Project** in the left sidebar, then click **Create**.
3. Enter Project Name: `mediahub`.
4. Path: Click Browse and select `/docker/mediahub`.
5. Source: Select **Create docker-compose.yml**, and paste the configuration below:

```yaml
version: "3.8"

services:
  mediahub-db:
    image: postgres:16-alpine
    container_name: mediahub-postgres
    restart: always
    environment:
      POSTGRES_DB: mediahub_db
      POSTGRES_USER: mediahub_user
      POSTGRES_PASSWORD: "ChangeMeToAStrongPassword123!" # <-- Set your password
      PGDATA: /var/lib/postgresql/data/pgdata
    ports:
      - "5432:5432"
    volumes:
      - /volume1/docker/mediahub/pgdata:/var/lib/postgresql/data/pgdata
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U mediahub_user -d mediahub_db"]
      interval: 10s
      timeout: 5s
      retries: 5
```

6. Click **Next** > verify the summary > check **Build image and start project** > click **Done**.
7. Container Manager will pull `postgres:16-alpine` and start the database. Verify that the status indicator turns green (**Healthy**).

3 Configure Cloudflare Zero Trust Tunnel on the NAS

This allows Cloudflare Workers and Vercel in the cloud to communicate with the NAS PostgreSQL instance securely over the web without requiring port forwarding.

Part A: Generate Tunnel in Cloudflare

1. Visit the Cloudflare Zero Trust Dashboard.
2. Go to **Networks** > **Tunnels** > Click **Create a tunnel**.
3. Select **Cloudflared**, name it `synology-mediahub`, and click **Save tunnel**.
4. Under **Choose your environment**, select **Docker**. Cloudflare will display a command with a tunnel token: `--token eyJhIjoi...` (copy this token).

Part B: Add Tunnel to Synology Container Manager

1. In DSM Container Manager, go to **Project** > click `mediahub` > **Action** > **Stop**.
2. Click **YAML Configurations** and add the `cloudflared` service right below PostgreSQL:

```yaml
  cloudflared:
    image: cloudflare/cloudflared:latest
    container_name: mediahub-tunnel
    restart: always
    command: tunnel run --token "YOUR_CLOUDFLARE_TUNNEL_TOKEN_HERE"
    depends_on:
      - mediahub-db
```

3. Click **Save** and restart the project. Cloudflare Dashboard will now show the tunnel status as **HEALTHY**.

Part C: Route Public Hostname to PostgreSQL

1. In Cloudflare Zero Trust > Tunnels > edit `synology-mediahub` > **Public Hostnames** > **Add a public hostname**.
2. Set **Subdomain:** `db` | **Domain:** `nushmedia.app`.
3. Set **Service:** Type = `TCP` | URL = `mediahub-db:5432`. Click **Save Hostname**.

4 Export Existing Data from Cloudflare / Vercel

Choose the method depending on where your current data resides:

Method A: Exporting from Vercel / Neon PostgreSQL

Run from PowerShell or terminal on your laptop:

```
pg_dump -d "postgres://USER:PASSWORD@ep-xyz.us-east-2.aws.neon.tech/neondb?sslmode=require"   --data-c
```

Method B: Exporting from Cloudflare D1 (Production)

Run using Wrangler CLI in your project directory:

```
npx wrangler d1 export inventory-tracker-db --remote --output=./d1_backup.sql
```

> **Fast Alternative:** You can also log into `https://nushmedia.app`, go to **Dashboard > Equipment**, and click **Export to Excel (.xlsx)** to get an instant, full spreadsheet of all inventory tags and serial numbers.

5 Import Data into the Synology PostgreSQL Database

Connect to the NAS database from your workstation and import the dump file:

```
psql -h <NAS_LOCAL_IP> -p 5432 -U mediahub_user -d mediahub_db -f mediahub_data_backup.sql
```

> **Automatic Schema Bootstrap:** MediaHub includes an auto-migrating schema engine in `lib/d1.ts` (`ensurePostgresSchema`). As soon as MediaHub connects, it will automatically create all required tables (`equipment` , `checkouts` , `nfc_cards` , `events` , `sop_documents` , `storage_map_layout`) and apply missing columns or indexes if they don't already exist.

6 Connect Cloudflare and Vercel to the Synology Database

Now configure the connection string across your cloud dashboards:

Connection String Format:

```
POSTGRES_URL="postgresql://mediahub_user:YourStrongPasswordHere@db.nushmedia.app:5432/mediahub_db?sslm
```

1. **On Vercel (vercel branch):**
   - Go to **Vercel Dashboard** > your project > **Settings** > **Environment Variables**.
   - Update `POSTGRES_URL` and `DATABASE_URL` to the connection string above.
   - Click **Deployments** > **Redeploy** to apply.
2. **On Cloudflare Workers (main branch):**
   - Go to **Cloudflare Dashboard** > **Workers & Pages** > `mediahub-website`.
   - Click **Settings** > **Variables and Secrets**.
   - Add / Update `POSTGRES_URL` with the connection string.
3. **On Local Development / NFC Kiosk Laptops:**
   - In your local `.env.local`, set: `POSTGRES_URL="postgresql://mediahub_user:YourStrongPasswordHere@<NAS_LOCAL_IP>:5432/mediahub_db?sslmode=disable"`
   - Local kiosks talk directly to the NAS at **< 1ms speed** over local WiFi!

7 Verification Checklist

- [ ] **Database Online:** Container Manager shows `mediahub-postgres` is healthy.
- [ ] **Tunnel Active:** Cloudflare Zero Trust shows `synology-mediahub` is active and sending heartbeats.
- [ ] **Live Checkouts:** Open `https://nushmedia.app/dashboard/checkout`, perform a test checkout, and verify the checkout record immediately appears in the database.
- [ ] **SOP Assistant:** Test `/dashboard/sop` to ensure documents load and the Gemini AI responds.
- [ ] **NFC Station:** Test the kiosk scanning feature to ensure instant response.

4. Recommended Ultimate Setup: Host the Web App on the NAS Too!

Since your Synology NAS is already running Docker, you can achieve maximum performance by **running both the Next.js web application container and PostgreSQL together on the NAS**.

> **Why this is the gold standard for school media clubs:**
>
> - **0ms Database Latency:** App and database communicate over `localhost`.
> - **Zero Cloud Subscription Limits:** No Vercel Pro fees ($20/mo/user), no Neon compute pauses.
> - **Offline Survival:** If the school's external internet connection drops during an auditorium event, members on local school WiFi can still load `http://<NAS_IP>:3000` to check out gear!
> - **Cloud Access:** The Cloudflare Tunnel routes `https://nushmedia.app` directly into port `3000` of the web container with free SSL and DDoS protection.

All-in-One `docker-compose.yml` for Synology NAS:

```yaml
version: "3.8"

services:
  # 1. PostgreSQL Database
  mediahub-db:
    image: postgres:16-alpine
    container_name: mediahub-postgres
    restart: always
    environment:
      POSTGRES_DB: mediahub_db
      POSTGRES_USER: mediahub_user
      POSTGRES_PASSWORD: "YourStrongPassword"
      PGDATA: /var/lib/postgresql/data/pgdata
    volumes:
      - /volume1/docker/mediahub/pgdata:/var/lib/postgresql/data/pgdata

  # 2. MediaHub Next.js Application Container
  mediahub-web:
    image: chaoseternity/mediahub:latest
    container_name: mediahub-web
    restart: always
    environment:
      POSTGRES_URL: "postgresql://mediahub_user:YourStrongPassword@mediahub-db:5432/mediahub_db"
      AUTH_URL: "https://nushmedia.app"
      AUTH_SECRET: "your-random-32-char-secret"
      GOOGLE_CLIENT_ID: "your-google-client-id"
      GOOGLE_CLIENT_SECRET: "your-google-client-secret"
      AZURE_AD_CLIENT_ID: "your-azure-client-id"
      AZURE_AD_CLIENT_SECRET: "your-azure-client-secret"
    ports:
      - "3000:3000"
    depends_on:
      - mediahub-db

  # 3. Cloudflare Tunnel (Exposes nushmedia.app securely to port 3000)
  cloudflared:
    image: cloudflare/cloudflared:latest
    container_name: mediahub-tunnel
    restart: always
    command: tunnel run --token "YOUR_CLOUDFLARE_TUNNEL_TOKEN"
    depends_on:
      - mediahub-web
```

Disaster Recovery & Backups: Because your database is stored in `/volume1/docker/mediahub/pgdata`, you can use Synology's built-in **Hyper Backup** or **Snapshot Replication** to take daily automated snapshots to a secondary drive, external USB, or Synology C2 Cloud. If anything ever corrupts, you can roll back your entire club inventory with 1 click!